LIMITI

1. Limiti di successioni

Verificare e giustificare i seguenti limiti

- $\lim_{n\to+\infty} \frac{n^n}{n!} = +\infty$
- $\lim_{n\to+\infty} \frac{n^n}{(3n)!} = 0$
- $\lim_{n\to+\infty} \frac{n^{4n}}{n!} = +\infty$
- $\lim_{n\to+\infty} \frac{n^{2n}}{(2n)!} = +\infty$
- $\lim_{n\to+\infty} \frac{n^{3n}}{(3n)!} = 0$
- $\lim_{n\to+\infty} \frac{n+4}{2n-2} = 1/2$
- $\lim_{n\to+\infty} \left(\frac{3n-\ln n}{n^3+\sin n}\right)^{1/2} = 0$
- $\lim_{n\to+\infty} n\sqrt{\frac{1}{n+1}} = +\infty$
- $\lim_{n\to+\infty} \frac{n+\ln n^2-e^n}{n^3-(\ln n)^4} = 0$
- $\lim_{n\to+\infty} \frac{n^2+(-1)^n}{2n^2-3(-1)^n} = 1/2$
- $\lim_{n\to+\infty} \frac{2^{n+1}-4^{n-1}}{3^n} = -\infty$
- $\lim_{n\to+\infty} \sqrt[n]{3n} = 1$
- $\lim_{n\to+\infty} \left(\frac{n^2+n}{n^2-n+1}\right)^{n^2} = +\infty$
- $\lim_{n\to+\infty} \left(\frac{n^2+n}{n^2-n+2}\right)^{n} = e^2$
- $\lim_{n\to+\infty} n\sin(3/n) = 3$
- $\lim_{n\to+\infty} \frac{2+\cos n}{\sin^2(1/n)} = +\infty$
- $\lim_{n\to+\infty} \frac{3^n(n^2+n+1)}{n^4(e^n+2e^{-n})} = +\infty$
- $\lim_{n\to+\infty} \frac{1}{3n^2}\sin^3(n^2) = 0$

2. Limiti di funzioni

Verificare e giustificare i seguenti limiti

- $\lim_{x\to0} \frac{\sin(3x)}{\arctan(5x)} = 3/5$
- $\lim_{x\to0} \frac{e^{\sin^3 x}-1}{\ln(1+x^2)} = 0$
- $\lim_{x\to0^+} \frac{\sqrt{x}-3\sqrt{x+x^4}}{x^2} = -\infty$
- $\lim_{x\to0^+} \frac{\sqrt{x}-\sqrt{x+x^4}}{x} = 0$
- $\lim_{x\to0^+} \frac{\sqrt{x}-\sqrt{x+x^2}}{x^2} = -\infty$
- $\lim_{x\to+\infty} \sqrt{x^2-3x}-x = -3/2$
- $\lim_{x\to0} \frac{xe^{\sin x}-x}{3\arctan^2 x} = 1/3$
- $\lim_{x\to+\infty} x^{1/x} = 1$
- $\lim_{x\to+\infty} \ln x - \sqrt{x} = -\infty$
- $\lim_{x\to+\infty} \frac{\ln\sqrt{x+1}}{x} = 0$
- $\lim_{x\to0} \frac{\ln\sqrt{x+1}}{x} = 1/2$
- $\lim_{x\to+\infty} \frac{x}{1-e^{2x}} = 0$
- $\lim_{x\to+\infty} x^2\Big(\ln(x^2+2)-2\ln x\Big) = 2$
- $\lim_{x\to+\infty} \left(\frac{x+2}{x+1}\right)^x = e$
- $\lim_{x\to0^+} x^x = 1$
- $\lim_{x\to0} \frac{\ln\cos x}{x^2} = -1/2$
- $\lim_{x\to+\infty} \sqrt{4x^2+1}-\sqrt{x^2-3} = +\infty$

- $\lim_{x\to+\infty} \frac{e^{2x}+e^x}{e^x+e^{x^2}} = 0$
- $\lim_{x\to+\infty} \sqrt{x^2+1} - \sqrt{x^2-\pi} = 0$
- $\lim_{x\to0^+} x^{\ln x} = +\infty$
- $\lim_{x\to0} \frac{\ln(1+x)+\ln(1-x)}{x^2} = -1$
- $\lim_{x\to0} x\sin(1/x) = 0$
- $\lim_{x\to+\infty} x\sin(1/x) = 1$
- $\lim_{x\to0} \frac{\ln(2x)}{\ln(5x)} = 1$
- $\lim_{x\to+\infty} \left(1+\frac{1}{\ln x}\right)^x = 1$
- $\lim_{x\to+\infty} \left(1+e^x\right)^{-x} = 0$
- $\lim_{x\to+\infty} \frac{\sin x}{\sqrt{x}+\cos x} = 0$
- $\lim_{x\to+\infty} \frac{x+\cos x}{3x-\sin x} = 1/3$
- $\lim_{x\to+\infty} \frac{5x^4-x^2\ln(1+x)-\sin x}{x^4-3x-\ln x} = 5$
- $\lim_{x\to0^+} \frac{1-\cos(2x)}{\ln(1-x)+\ln(1+x)} = -2$
- $\lim_{x\to0} \frac{1}{2}\frac{\ln(1+x^2)}{1-\cos x} = 1$
- $\lim_{x\to+\infty} \left(\frac{\ln x}{x}\right)^{1/x} = 1$
- $\lim_{x\to0^+} \frac{x^2-x\ln x}{\sin(\arctan x)} = +\infty$
- $\lim_{x\to0} \frac{(1-3x)^{1/x}}{x} = +\infty$
- $\lim_{x\to0} \frac{\sqrt{x^2+\arctan^2 x}}{x+x^2} = \sqrt{2}$